Evaluation of Seminal Fluid Analysis among Men Experiencing Immunological Infertility in South-West Nigeria

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ABSTRACT

Immunological infertility is an important but under-investigated cause of male infertility, particularly with respect to antisperm antibodies (ASA) and their relationship with seminal fluid characteristics. This study compares serum ASA levels and seminal fluid parameters in men with immunological infertility and fertile controls. The case–control study involved 32 men each experiencing infertility and fertile controls, recruited from South-West Nigeria using purposive sampling. Seminal fluid was analysed and serum ASA concentration quantified using enzyme-linked immunosorbent assay. Data were analysed in R; Fisher's exact test, the Shapiro–Wilk test, the Mann–Whitney U test, and Spearman's rank correlation were used as appropriate, with significance set at $p < 0.05$. Serum ASA levels were significantly higher among men experiencing immunological infertility than fertile controls. Significant differences between group were observed in semen volume, total motility, and normal sperm morphology, whereas progressive motility and sperm concentration did not differ significantly. Although most seminal fluid parameters among men experiencing immunological infertility met or exceeded the WHO 2021 lower reference values, several were nonetheless significantly lower than those of fertile controls. Serum ASA levels showed significant negative correlations with total motility ($r = -0.18$), normal sperm morphology ($r = -0.31$), and sperm concentration ($r = -0.14$). Serum ASA levels are associated with selected seminal fluid characteristics among men with immunological infertility, even when conventional semen parameters remain within WHO reference limits. These findings support incorporating immunological assessment alongside conventional seminal fluid analysis in the investigation of male infertility.

Key words: Antisperm antibodies; Immunological infertility; Seminal fluid analysis; Male infertility; ELISA

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1. INTRODUCTION

Infertility is a major global reproductive health problem, and its social, psychological, and medical implications for couples continue to grow (1). Although reproductive medicine has advanced considerably, many cases still present diagnostic challenges, particularly when no apparent cause emerges from typical clinical assessment. Male factors contribute significantly to couple infertility, yet in many developing countries, including Nigeria, clinical attention remains disproportionately focused on female-related factors (2). This skewed perception has been associated with inadequate assessment of men during infertility investigation and reduced awareness of major male reproductive disorders such as immunological infertility.

Immunological infertility arises from the production of antisperm antibodies that

adversely affect sperm function and fertility potential, yet it is often overlooked during routine infertility investigation (3,4). Although seminal fluid analysis remains the cornerstone of male infertility investigation, it may not on its own reveal an underlying immunological cause, potentially leading to missed or delayed diagnosis.

The conventional Nigerian approach to infertility investigation is largely limited to semen analysis and hormonal assessment; immunological assessment is not routinely performed. Consequently, some men with immunological infertility remain undiagnosed or are managed incorrectly, becoming chronically infertile, undergoing repeated failed treatment, and experiencing considerable emotional and financial burden.

Although evidence increasingly links immunological infertility with altered seminal fluid characteristics, little is known about this relationship among South-West Nigerian men specifically. Most available studies address male infertility broadly, and few case–control studies compare men with immunological infertility directly against fertile men in this region, limiting evidence-based diagnosis and management. This study was therefore undertaken to address this gap and to strengthen the laboratory diagnosis and clinical management of immunological infertility among men in South-West Nigeria..

2. MATERIALS AND METHODS

2.1. Study area: This study was conducted in South-West Nigeria, comprising the states of Lagos, Ogun, Oyo, Osun, Ondo, and Ekiti.

2.2. Study design: A case–control design was used to compare men experiencing immunological infertility with fertile controls in South-West Nigeria.

2.3. Study population: The study population comprised male participants from the six South-West Nigerian states, consisting of men with immunological infertility and fertile men recruited as controls.

2.4. Inclusion criteria: Men aged 20–60 years experiencing unexplained infertility (cases). Men aged 20–60 years with two or more biological children (controls). Willingness to provide informed consent.

2.5. Exclusion criteria: Refusal or inability to provide informed consent. Age under 20 or over 60 years. Disease conditions likely to impact reproductive function.

2.6. Sampling technique and sample size: Participants were recruited using a purposive sampling technique. A total of 64 men were enrolled: 32 experiencing infertility and 32 fertile controls, giving a 1:1 case-to-control ratio. Sample size was calculated and validated using OpenEpi (version 3.01) at 80% statistical power.

2.7. Ethical considerations: Ethical approval for this study was obtained from the Institutional Review Board of the Nigerian Institute of Medical Research (NIMR; Project number: IRB/23/039). Written informed consent was obtained from all participants after a full explanation of the study's aims, procedures, and their right to withdraw at any time without consequence, prior to specimen collection. The study was conducted in accordance with the Declaration of Helsinki.

2.8. Specimen collection: Two biological specimens were collected from each participant: a semen sample and a venous blood sample. Specimen collection followed questionnaire administration and confirmation of informed consent. Semen samples were produced by masturbation into a sterile, wide-mouthed container after an abstinence period of 3–6 days and were allowed to liquefy at room temperature before analysis.

2.9. Semen analysis: Semen volume, sperm concentration, total and progressive motility, and normal sperm morphology were assessed according to the WHO laboratory manual for

the examination and processing of human semen, 6th edition (2021). Morphology was assessed using strict (Kruger) criteria.

2.10. Antisperm antibody assay: Venous blood was collected, and serum was separated and stored at -20°C until analysis. Serum antisperm antibody concentrations were determined by enzyme-linked immunosorbent assay (ELISA) produced by Eiaere Medical Technology Co., Ltd with sensitivity of 1.19ng/ml.

2.11. Statistical analysis: Data were analysed using R statistical software. Continuous variables were summarised as mean ± standard deviation or median (interquartile range) according to distribution, and categorical variables as frequencies and percentages. Between-group comparisons of continuous variables were performed using the Mann–Whitney U test; categorical variables were compared using Fisher's exact test with Monte Carlo simulation (10,000 resamples). Grubbs' test was used to screen for extreme outliers. Receiver operating characteristic (ROC) curve analysis was used to derive the optimal antisperm antibody cut-off value (Youden index), and the association between antisperm antibody positivity and infertility was quantified using the odds ratio with 95% confidence interval. Statistical significance was set at $p < .05$.

3. RESULTS

Characteristics of study participants: The study included 64 participants: 32 men experiencing infertility and 32 fertile controls. Demographic information; age group, educational level, religion, self-reported medical conditions, years in marriage, and number of children with a partner was available for all participants and is presented in Table 1.

The distribution of participants across age groups was comparable between the two study groups ($p = 0.728$), as were educational level ($p = 0.394$), religion ($p = 0.750$), self-reported medical conditions ($p = 1.000$), and years in marriage ($p = 0.241$). The number of children with a partner differed significantly between groups ($p < 0.001$), with fertile controls having a higher median than men experiencing infertility, a difference consistent with the definition of the study groups rather than an unexpected imbalance.

Descriptive statistics of serum antisperm antibody levels: Men experiencing immunological infertility had a higher mean serum ASA concentration (6.45 ng/mL) than fertile controls (3.59 ng/mL), together with considerably greater variability, as reflected in the larger standard deviation (6.94 ng/mL) and wider range. The marked gap between the mean (6.45 ng/mL) and median (3.48 ng/mL), together with a maximum value of 38.30 ng/mL, suggested the presence of an extreme observation capable of disproportionately influencing summary statistics and subsequent inferential analyses. Formal outlier assessment was therefore undertaken before between-group comparison.

Detection and treatment of an extreme asa value: Serum ASA concentrations from the 32 participants experiencing immunological infertility were screened for extreme values before comparative analysis. The upper IQR fence was calculated as $Q3 + 1.5 \times IQR = 6.13 + (1.5 \times 3.00) = 10.63$ ng/mL. Four ASA concentrations exceeded this fence (13.07, 15.54, 16.77, and 38.30 ng/mL); because 38.30 ng/mL was markedly separated from the other observations, it was subjected to formal testing using Grubbs' test.

An observation was considered a statistically significant extreme outlier when the Grubbs' test statistic exceeded the critical value at the specified significance level.

The value of 38.30 ng/mL exceeded the Grubbs' critical value of 3.26 and was confirmed as a statistically significant extreme outlier ($G = 4.59$, $p < .0001$). The participant's source records were reviewed to confirm the observation was not attributable to data-entry or analytical error; following this verification,

the value was excluded from subsequent analysis. The three remaining values above the IQR fence were retained, as they did not meet the statistical criterion for exclusion. The final analytical sample therefore comprised 31 participants experiencing immunological infertility and 32 fertile controls.

Table 1: Socio-demographic characteristics of study participants

Characteristic	Control (n = 32)	Test (n = 32)	p-value
Age group (years)			**0.728[a]**
20–30	1 (3.1%)	2 (6.3%)	
31–40	10 (31.3%)	11 (34.4%)	
41–50	15 (46.9%)	16 (50.0%)	
>51	6 (18.8%)	3 (9.4%)	
Educational level			0.394[a]
SSCE	4 (12.5%)	3 (9.4%)	
OND	2 (6.3%)	4 (12.5%)	
BSc	10 (31.3%)	15 (46.9%)	
MSc	14 (43.8%)	10 (31.3%)	
PhD	2 (6.3%)	0 (0.0%)	
Religion			0.750[a]
Christian	25 (78.1%)	27 (84.4%)	
Muslim	7 (21.9%)	5 (15.6%)	
Self-reported medical condition			1.000[a]
None	31 (96.9%)	31 (96.9%)	
Reported	1 (3.1%)	1 (3.1%)	
Years in marriage, median (IQR)	10.0 (6.5)	8.0 (3.0)	0.241[b]
Children with partner, median (IQR)	3.0 (1.0)	0.0 (0.0)	<0.001[b]

Categorical data are presented as frequency and percentage, n (%). Continuous variables are presented as median (IQR). (a) Fisher's exact test with Monte Carlo simulation; (b) Mann–Whitney U test. Statistical significance was set at $p < 0.05$.

Table 2: Descriptive statistics of serum ASA concentrations by study group, prior to outlier assessment

Study **group**	n	Mean (ng/mL)	SD	Median	IQR	Min	Max
Fertile control	32	3.59	1.28	3.13	0.88	2.01	6.98
Test (immunological infertility)	32	6.45	6.94	3.48	3.00	1.99	38.30

ASA = antisperm antibody; SD = standard deviation; IQR = interquartile range. Values are based on the original dataset prior to outlier assessment.

Table 3: Summary of extreme-value screening and Grubbs' test for serum ASA concentration

Parameter	Result
Initial sample size (n)	32
IQR upper fence	10.63 ng/mL
Number of values above the upper fence	4
Extreme ASA value assessed	38.30 ng/mL (Participant 45)
Grubbs' test statistic (G)	4.59
Critical value	3.26
p-value	< .0001
Decision	Significant extreme outlier detected
Final sample size after exclusion	31

Table 4: Comparison of serum antisperm antibody levels between study groups, following outlier exclusion

Study group	n	Mean (ng/mL)	SD	Median	IQR	Min	Max
Fertile control	32	3.59	1.28	3.13	0.88	2.01	6.98
Test (immunological infertility)	31	5.42	3.85	3.34	2.92	1.99	16.77

Mann–Whitney U = 348, p = .042. ASA = antisperm antibody; SD = standard deviation; IQR = interquartile range. Statistical significance was set at $p < 0.05$.

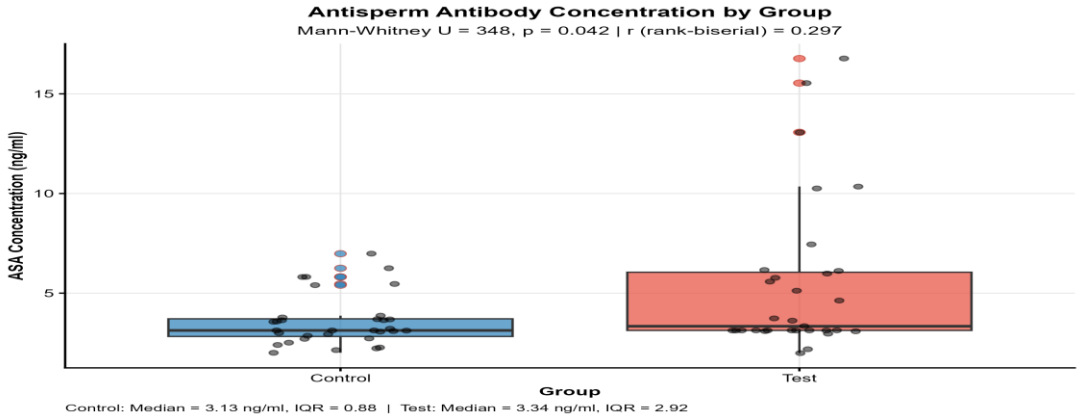


Figure 1: Serum antisperm antibody (ASA) concentration by study group

Comparison of Serum ASA Levels Between Study Groups

The median serum ASA concentration was higher among men experiencing immunological infertility than among fertile controls (3.34 vs 3.13 ng/mL), with greater variability in the infertility group as reflected in its higher standard deviation and wider interquartile range. This difference was statistically significant (U = 348, p = .042).

Boxplot showing the distribution of serum ASA concentrations among fertile controls and men experiencing immunological infertility. The horizontal line within each box represents

the median, the box represents the interquartile range (IQR), and the whiskers extend to the most extreme observations within 1.5 × IQR. The figure represents the dataset after exclusion of the confirmed extreme value.

Consistent with Table 4, the boxplot shows a slightly higher median serum ASA concentration among men experiencing immunological infertility (3.34 ng/mL) than among fertile controls (3.13 ng/mL), together with the greater variability and wider distribution of values in the infertility group compared with the relatively compact control distribution.

Comparison of seminal fluid parameters between study groups: The seminal fluid parameters assessed were semen volume, total motility, progressive motility, normal sperm morphology, and sperm concentration. Following assessment of data distribution using the Shapiro–Wilk test, the Mann–Whitney U test was used to compare parameters between groups; results are presented in Table 5.

Three of the five seminal fluid parameters differed significantly between groups. Semen volume was significantly lower among men experiencing immunological infertility than fertile controls (2.68 ± 1.32 mL vs 3.40 ± 1.41 mL; U = 656, p = .027), as was total motility (59.35 ± 7.93% vs 71.25 ± 13.56%; U = 770, p = .0001). The most pronounced difference was in normal sperm morphology: 25.48 ± 4.15% among men experiencing immunological infertility versus 73.44 ± 12.92% among fertile controls (U = 992, p < .0001), with a rank-biserial correlation of -1.000, indicating complete separation between the two groups on this parameter, the ranges of the two groups did not overlap at all.

No statistically significant difference was observed in progressive motility (49.22 ± 13.51% vs 43.55 ± 7.09%; U = 600, p = .147) or sperm concentration (69.34 ± 48.38 vs 52.77 ± 42.60 × 10^6/mL; U = 618, p = .096). Overall, significant between-group differences were confined to semen volume, total motility, and normal sperm morphology.

Comparison of seminal fluid parameters with WHO 2021 lower reference values: To further contextualise the seminal fluid findings, parameters among men experiencing immunological infertility were compared with the lower reference values in the WHO 2021 (6th edition) manual.

The majority of participants had seminal fluid parameters at or above the WHO 2021 lower reference limits: 96.8% for semen volume, total motility, and sperm concentration, 93.5% for progressive motility, and 100.0% for normal morphology. Despite the statistically significant between-group differences observed in semen volume, total motility, and normal sperm morphology, most men in the infertility group would not be classified as infertile on conventional WHO semen criteria alone, highlightingg the potential diagnostic contribution of antisperm antibody assessment in this population.

Relationship between serum ASA levels and seminal fluid parameters: The relationship between serum ASA levels and seminal fluid parameters among men experiencing immunological infertility was assessed using Spearman's rank correlation, examining ASA against semen volume, total motility, progressive motility, normal sperm morphology, and sperm concentration.

Serum ASA levels were negatively correlated with all five seminal fluid parameters assessed, though the strength and significance of these relationships varied. Statistically significant negative relationships were observed with total motility (r = -0.18, p = .041), normal sperm morphology (r = -0.31, p = .001), and sperm concentration (r = -0.14, p = .048), indicating that higher ASA levels were associated with lower values on these parameters. The relationships with semen volume (r = -0.04, p = .827) and progressive motility (r = -0.16, p = .068) were not statistically significant.

Overall, serum ASA levels were significantly related to some, but not all, seminal fluid parameters. The strongest correlation was with normal sperm morphology (r = -0.31), followed by total motility and sperm concentration, all weak in magnitude but statistically significant.

Table 5: Comparison of seminal fluid parameters between study groups

Seminal fluid parameter	Control (mean ± SD)	Test (mean ± SD)	U	p-value	Rank-biserial r
Semen volume (mL)	3.40 ± 1.41	2.68 ± 1.32	656	.027	-0.322
Total motility (%)	71.25 ± 13.56	59.35 ± 7.93	770	.0001	-0.552
Progressive motility (%)	49.22 ± 13.51	43.55 ± 7.09	600	.147	-0.209
Normal morphology (%)	73.44 ± 12.92	25.48 ± 4.15	992	< .0001	-1.000
Sperm concentration (×10⁶/mL)	69.34 ± 48.38	52.77 ± 42.60	618	.096	-0.246

SD = standard deviation; U = Mann–Whitney U test statistic; r = rank-biserial correlation effect size. All p-values are two-tailed. Statistical significance was set at $p < 0.05$.

Table 6: Comparison of seminal fluid parameters among men experiencing immunological infertility with WHO 2021 lower reference values

Parameter	Mean	SD	Median	WHO 2021 lower reference value	Below reference, n (%)	At/above reference, n (%)
Semen volume (mL)	2.68	1.32	2.50	≥ 1.4	1 (3.2)	30 (96.8)
Total motility (%)	59.38	7.80	60.00	≥ 42	1 (3.2)	30 (96.8)
Progressive motility (%)	43.44	7.01	45.00	≥ 30	2 (6.5)	29 (93.5)
Sperm concentration (×10⁶/mL)	52.77	42.60	31.00	≥ 16	1 (3.2)	30 (96.8)
Normal morphology (%)	25.48	4.15	25.00	≥ 4	0 (0.0)	31 (100.0)

Percentages are calculated within the immunological infertility group (n = 31).

DISCUSSION

The demographic and background characteristics of participants were generally comparable between groups, with no significant differences in age group, educational level, religion, self-reported medical conditions, or years in marriage. This comparability strengthens confidence that the laboratory findings reported below are not confounded by differences in these background characteristics. The significant difference in number of children with a partner was expected, reflecting the criteria used to define

the study groups rather than an unexpected imbalance.

Serum Antisperm Antibody Levels Among Study Participants

A statistically significant difference was observed in serum ASA levels between men experiencing immunological infertility and fertile controls. Following exclusion of one statistically confirmed extreme value, the infertility group retained a higher median serum ASA concentration than controls, supporting the relevance of antisperm antibodies in the laboratory investigation of male immunological infertility.

Table 7: Relationship between serum ASA levels and seminal fluid parameters among men experiencing immunological infertility

Seminal fluid parameter	r (Spearman)	Direction and strength	p-value
Semen volume	-0.04	Very weak negative	0.827
Total motility	-0.18	Very weak negative	0.041
Progressive motility	-0.16	Very weak negative	0.068
Normal sperm morphology	-0.31	Weak negative	0.001
Sperm concentration	-0.14	Very weak negative	0.048

r = Spearman's rank correlation coefficient. Statistical significance was set at $p < 0.05$..

This finding is consistent with the literature indicating that antisperm antibodies are associated with male infertility, with occurrence and clinical significance varying across populations (5). Antisperm antibody development has been associated with conditions that disrupt normal protection of sperm antigens from immune recognition, including infection, inflammation, trauma, surgical procedures, and disruption of the blood–testis barrier (3,4).

The present finding is also consistent with literature reporting a higher occurrence of antisperm antibodies among infertile men, while noting considerable variation in reported prevalence between populations and studies(6). Differences in study population, criteria used to define infertility, antibody detection methods, and laboratory procedures have been proposed as possible explanations for this variation (4,7).

The higher serum ASA levels observed among men experiencing immunological infertility are also relevant to laboratory diagnosis. Although ELISA provides a quantitative measure of antibody concentration, concentration alone does not necessarily establish an antibody's functional effect on spermatozoa; ASA findings should therefore be interpreted alongside seminal fluid analysis and other relevant clinical or laboratory information (8).

Comparison of Seminal Fluid Parameters Between Study Groups

Statistically significant differences between groups were observed in semen volume, total motility, and normal sperm morphology, while progressive motility and sperm concentration did not differ significantly. This between-group comparison addresses a distinct question from the correlation analysis reported subsequently: whether a parameter differs between groups is separate from whether variation in ASA levels is related to variation in that parameter within the infertility group. A parameter may therefore differ significantly between groups without showing a significant correlation with ASA levels within the infertility group, and vice versa.

Semen volume was significantly lower among men experiencing immunological infertility, although the WHO 2021 comparison showed that most participants nonetheless had volumes at or above the applicable lower reference value, indicating a relative difference between groups rather than widespread sub-reference values within the infertility group.

Total motility was also significantly lower among men experiencing immunological infertility, consistent with the literature identifying reduced motility as one of the seminal fluid abnormalities most frequently associated with antisperm antibodies; antibody attachment to spermatozoa may interfere with movement and with sperm transport and fertilisation (4,8,9). This finding agrees with studies reporting significantly reduced motility among ASA-positive men, including those discussed by Verón et al. (11) and Gupta et al. (4). However, most participants in the infertility group still had total motility values at or above the WHO 2021 lower reference value, so the result reflects a significant relative difference between groups rather than widespread values falling below the absolute reference threshold.

The most pronounced difference between groups was in normal sperm morphology, which was substantially lower among men experiencing immunological infertility and represented the most clearly distinguishing seminal fluid characteristic in this study. This is compatible with literature describing possible associations between antisperm antibodies and abnormal morphology, with immune-mediated inflammatory and antibody-mediated effects proposed as possible mechanisms (9). The literature is not entirely consistent on this point, however, with differences in study populations, morphology assessment criteria, sample size, and antibody detection methods suggested as possible explanations for variation between studies (11).

Progressive motility did not differ significantly between groups; although the infertility group had a lower mean value, this did not reach statistical significance. This differs from some previous studies reporting reduced motility among ASA-positive men, which may reflect differences in study design, population, laboratory methods, or the nature of ASA assessment across studies.

Sperm concentration was also lower, but not significantly, among men experiencing immunological infertility. This is consistent with variable evidence in the literature: Cui et al. (10), in a systematic review and meta-analysis, reported poorer semen parameters, including lower sperm concentration, among men with ASA, while the broader literature acknowledges inconsistent findings on this specific association.

Overall, between-group differences were most pronounced for normal sperm morphology and total motility, with semen volume also differing significantly; progressive motility and sperm concentration did not differ significantly.

Comparison with WHO 2021 Lower Reference Values

Comparison with WHO 2021 lower reference values provided important clinical context: although statistically significant between-group differences were observed for some parameters, most participants in the infertility group had individual values at or above the WHO 2021 lower reference limits. This indicates that meeting an established reference threshold does not necessarily equate to seminal fluid characteristics comparable with those of fertile men, and that the men in the case group would largely not be classified as infertile on conventional semen criteria alone, a pattern that reinforces the potential diagnostic value of antisperm antibody assessment in cases of otherwise unexplained infertility. The direct between-group comparison and the WHO reference-value comparison address distinct questions and are best interpreted together rather than in isolation.

Relationship Between Serum ASA Levels and Seminal Fluid Parameters

The correlation analysis demonstrated negative relationships between serum ASA levels and all five seminal fluid parameters, though of varying strength and significance. Statistically significant negative relationships were observed with total motility, normal sperm morphology, and sperm concentration, whereas the relationships with semen volume and progressive motility were not significant.

The significant negative relationship with total motility is consistent with the literature; proposed mechanisms include antibody interference with sperm membrane function, calcium signalling, energy metabolism, and transport through the female reproductive tract (4,8,9). Although the correlation observed here was weak in magnitude, its statistical significance indicates that higher serum ASA levels were associated with lower total motility within the infertility group, consistent with prior reports of reduced motility among ASA-positive men (4,10).

The strongest of the five ASA–seminal fluid correlations was with normal sperm morphology. This is compatible with literature describing possible immune-mediated inflammatory and antibody-mediated effects on sperm structure (9), although Verón et al. (11) note that this relationship may vary according to population, morphology assessment criteria, and antibody detection method. The present finding therefore adds to the evidence supporting an ASA–morphology association, while its weak magnitude indicates that other factors also contribute to variation in morphology.

A significant negative relationship was also observed with sperm concentration, consistent with Cui et al.'s (10) systematic review and meta-analysis reporting poorer semen parameters, including lower concentration, among ASA-positive men, although the literature as a whole has not been entirely consistent on this association.

The relationships with semen volume and progressive motility were not statistically significant, indicating that the ASA–seminal fluid relationship is not uniform across all parameters. Notably, the absence of a significant ASA–semen volume correlation means the significant between-group difference in semen volume reported above should not be interpreted as evidence that ASA concentration was responsible for that difference, the group comparison and the correlation analysis address different relationships and should not be conflated.

Overall, the correlation findings show that serum ASA levels were significantly associated with total motility, normal sperm morphology, and sperm concentration, but not with semen volume or progressive motility, reinforcing the multifactorial nature of male infertility and the value of assessing serum ASA alongside conventional seminal fluid analysis.

5. CONCLUSION

This study set out to determine and compare serum antisperm antibody levels between men experiencing immunological infertility and fertile controls, and to examine their relationship with seminal fluid parameters. A statistically significant difference in serum ASA levels was found between groups, with higher levels among men experiencing immunological infertility.

The study also identified significant differences in selected seminal fluid parameters between groups and significant relationships between serum ASA levels and total motility, normal sperm morphology, and sperm concentration. Although most seminal fluid parameters among men experiencing immunological infertility met or exceeded the WHO 2021 lower reference values, several were nonetheless significantly lower than those of fertile controls, indicating that meeting a reference threshold does not guarantee seminal fluid quality equivalent to that of fertile men.

Overall, these findings indicate that serum antisperm antibody levels are associated with selected seminal fluid characteristics among men experiencing immunological infertility and support the consideration of immunological assessment alongside conventional seminal fluid analysis in the investigation of male infertility.

Based on the finidings, the following recommendations are made;

1. Clinicians and fertility laboratories should consider immunological assessment as part of male infertility investigation, since this study demonstrated significant relationships between serum ASA levels and selected seminal fluid parameters even though most measured parameters among men experiencing immunological infertility were at or above the WHO 2021 lower reference values.

2. Antisperm antibody results should be interpreted alongside seminal fluid findings rather than as an isolated measurement, since the relationship between serum ASA levels and seminal fluid parameters was not uniform across all parameters assessed.

3. Further studies with larger sample sizes and participants from other regions of Nigeria are needed to corroborate the relationship between serum ASA levels and seminal fluid parameters.

4. Future research should consider complementary antisperm antibody assessment methods, including assays that evaluate antibodies bound directly to spermatozoa, to clarify the functional significance of detected antibodies.

5. Longitudinal and larger-scale studies are recommended to further investigate the clinical significance and consistency of the ASA–seminal fluid relationship across different populations.

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REFERENCES

1. World Health Organization. Infertility. Geneva: WHO; 2023. Available from: https://www.who.int/news-room/fact-sheets/detail/infertility
2. Agarwal A, Baskaran S, Parekh N, Cho CL, Henkel R, Vij S, Arafa M, Selvam MKP, Shah R. Male infertility. *Lancet*. 2021;397(10271):319–333.
3. Leathersich S, Hart RJ. Immune infertility in men. *Fertil Steril.* 2022;117(6):1120–1131.
4. Gupta S, Sharma R, Agarwal A, Boitrelle F, Finelli R, Farkouh A, et al. Antisperm antibody testing: a comprehensive review of its role in the management of immunological male infertility and results of a global survey of clinical practices. *World J Mens Health*, 2022;40(3):380–398.
5. Adejumo BI, Oke AA, Emokpae MA, Erhobor TA, Agba MI, Oni A. A study of antisperm antibody among infertile subjects investigated for infertility in Lagos, Nigeria. *Ann Biomed Sci.*, 2014;13:29–35.
6. Shiraishi Y, Shibahara H, Koriyama J, Kikuchi K, Hirano Y, and Suzuki M. Clinical and biological characteristics of immunologically infertile men with sperm immobilizing antibody. *Fertility and Sterility*, 2008; 90, S96.

7. Mazumdar S, Levine AS. Antisperm antibodies: etiology, pathogenesis, diagnosis, and treatment. *Fertil Steril,* 1998;70(5):799–810.
8. Silva AFN, Ramalho-Santos J, Amaral S. The impact of antisperm antibodies on human male reproductive function: an update. *Reproduction,* 2021; 162(4):R55–R69.
9. Bohring C, Krause W. The role of antisperm antibodies during fertilization and for immunological infertility. *Am J Reprod Immunol.,* 2005; 54(5):279–288.
10. Cui D, Han G, Shang Y, Liu C, Xia L, Li L, Yi S. Antisperm antibodies in infertile men and their effect on semen parameters: a systematic review and meta-analysis. *Clin Chim Acta*, 2015; 444:29–36.
11. Verón GL, Molina RI, Tissera AD, Estofan GM, Marin-Briggiler CI, Vazquez-Levin MH. Incidence of sperm surface autoantibodies and relationship with routine semen parameters and sperm kinematics. *Am J. Reprod Immunol.,* 2016;76(1):59–69.